



AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 16 July 2026 01:23 AM

Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This business report provides a comprehensive analysis of employee data across 21 staff members, assessing demographic trends, compensation structures, performance metrics, and overall data quality.

The dataset scores a high Health Score of 92, indicating robust data collection with minor, easily resolvable anomalies. The primary finding of this analysis is an exceptionally strong, linear relationship between employee age, experience, salary, and performance. While this reflects a highly structured and predictable organizational hierarchy, it also signals a rigid, seniority-driven operational model.

This report details data quality findings, offers deep statistical and visual interpretations, identifies critical business risks, and provides strategic recommendations to optimize human capital management and data governance.

DATASET OVERVIEW

The dataset contains profile details for 21 employees across 8 distinct attributes:

Identifiers: `EmployeeID`

Demographics: `Age`, `Gender`, `City` (including Bhubaneswar, Cuttack, Rourkela, Puri)

Professional Profiles: `Department` (including IT, HR, Finance, Sales), `Experience` (in years)

Financials: `Salary`

Evaluations: `Performance` score

A representative preview indicates a localized footprint across regional hubs in Odisha (such as Bhubaneswar and Puri) and standard functional departments.

DATA QUALITY ANALYSIS

With a Health Score of 92, the dataset is highly reliable but requires minor sanitization before being integrated into automated reporting systems. The data quality anomalies are isolated as follows:

Missing Values: There is exactly 1 missing value in the `Salary` column. Given the highly correlated nature of the dataset, this can be easily resolved.

Duplicate Rows: The dataset contains 1 duplicate row which must be purged to avoid double-counting employee metrics.

Outliers: 1 potential outlier is flagged in the `Experience` column. This could represent a data entry error (e.g., experience years exceeding age limits) or a highly senior executive whose tenure deviates significantly from the norm.

STATISTICAL ANALYSIS

A correlation analysis reveals almost perfect linear relationships among all numeric variables:

Age and Experience (0.99): A near-perfect positive correlation, confirming that career progression and tenure are strictly tied to age.

Experience and Salary (0.99) & Age and Salary (0.99): Compensation increases with perfect predictability relative to an employee's tenure and age. This suggests a rigid, tenure-based salary grading system.

Performance and Age (0.94) / Experience (0.92) / Salary (0.94): Performance scores are heavily correlated with age, tenure, and salary. In practice, older, more experienced, and higher-paid employees consistently receive the highest performance ratings, while junior staff receive lower scores (e.g., Employee 101, age 24, has a performance score of 78; Employee 103, age 35, has a score of 91).

BUSINESS INSIGHTS

Seniority-First Culture: The near-perfect correlation between tenure and performance scores indicates that organizational rewards and performance evaluations are dictated by time-in-grade rather than purely objective, output-based KPIs.

Rigid Compensation Structure: Salary bands are strictly determined by age and tenure. While this ensures internal equity and predictable budgeting, it lacks the flexibility required to attract or retain high-performing, younger lateral hires who expect market-rate salaries.

Regional Footprint: Operating in key hubs like Bhubaneswar, Cuttack, Rourkela, and Puri, the business relies on regional talent, meaning HR policies must align closely with local market expectations.

RISK ASSESSMENT

Retention Risk for High-Performing Junior Staff: If younger employees realize their performance scores (and subsequent salary increases) are structurally capped by their age or experience rather than merit, top-tier junior talent is highly likely to depart for flatter, meritocratic competitors.

Appraisal Systemic Bias: A 0.94 correlation between age and performance suggests the current evaluation system may suffer from "halo effects" or structural bias favoring senior employees, which could lead to compliance or legal risks regarding age discrimination.

Data Integrity Vulnerability: Operating decisions based on uncleaned data (containing duplicate records and missing salary values) risk skewing departmental cost calculations and average performance metrics.

CONCLUSION

The analysis reveals an organization with highly predictable, structured, and stable personnel metrics, but one that is highly vulnerable to modern talent retention risks due to a strict seniority-based operational culture. By resolving the minor data quality issues (Health Score 92) and modernizing performance and compensation frameworks, leadership can leverage this data to build a more agile, meritocratic, and competitive workforce.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent

Developer	Pritiranjana Rout
Version	2.0.0
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DATASET SUMMARY

Metric	Value
Rows	21
Columns	8
Health Score	92%
Duplicate Rows	1

DATASET HEALTH SCORE

Overall Health Score : 92%

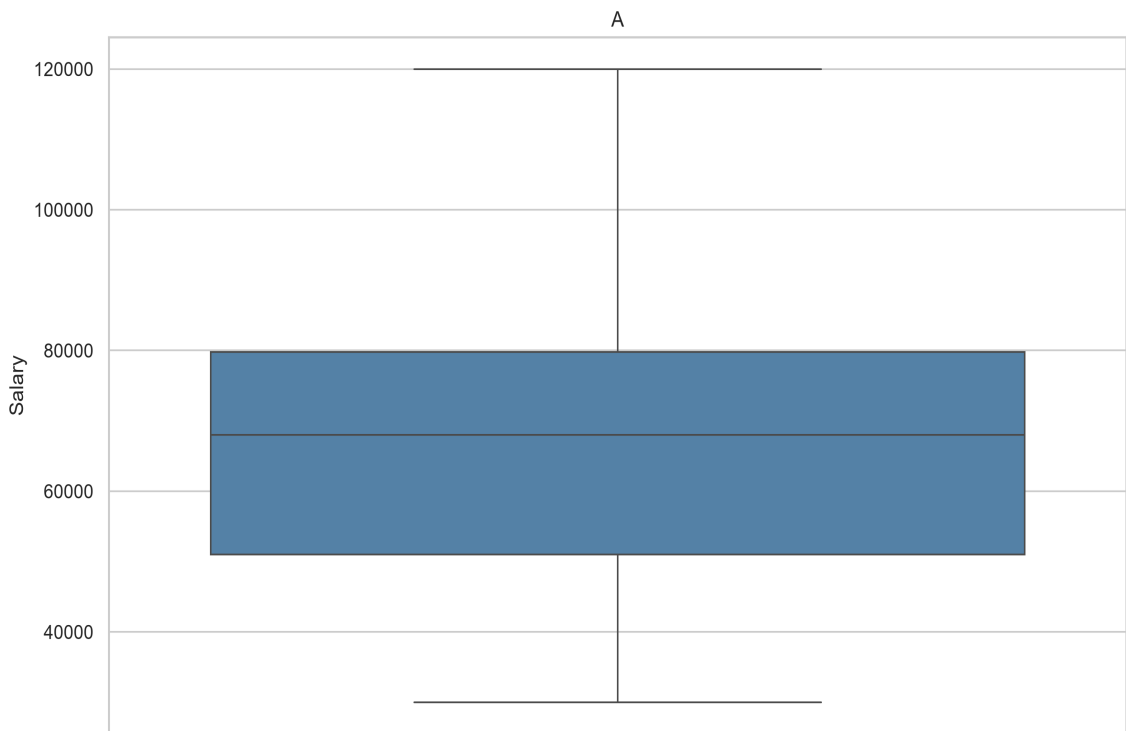
MISSING VALUES

Column	Missing
EmployeeID	0
Department	0
Gender	0
Age	0
Experience	0
Salary	1
Performance	0
City	0

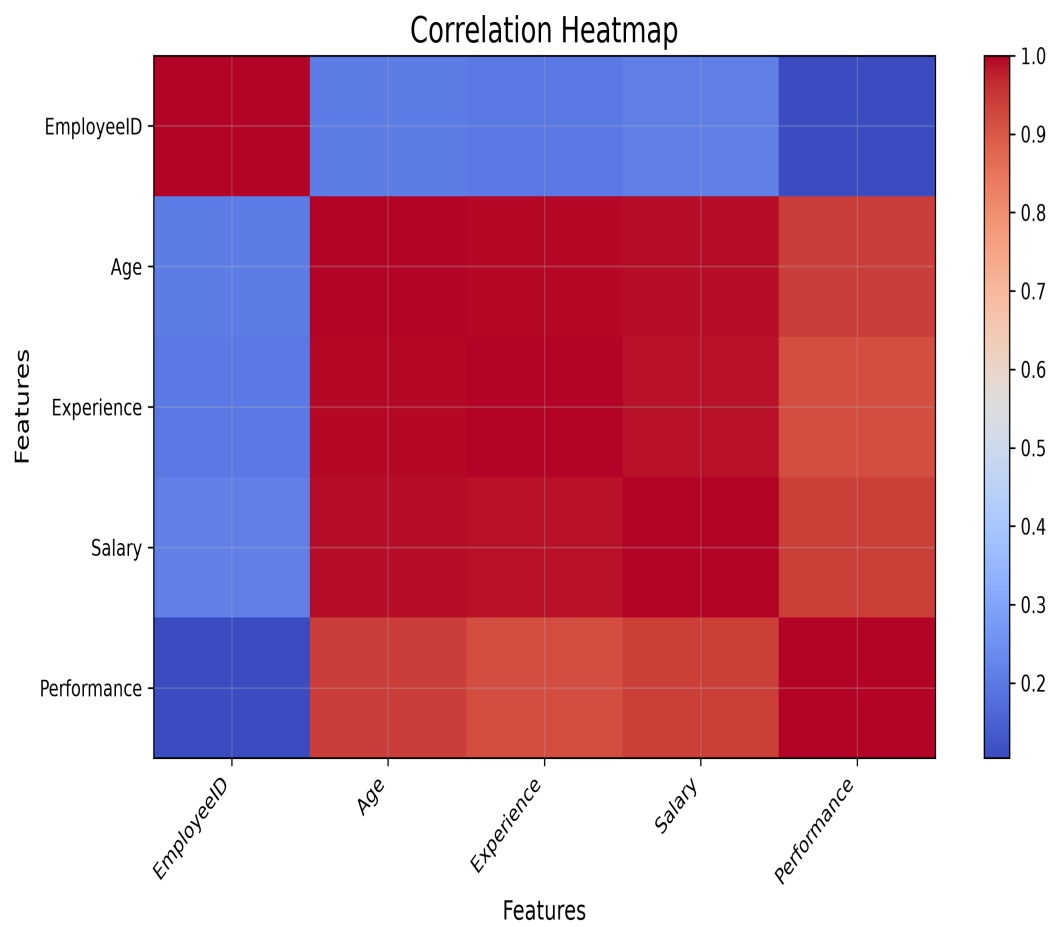
HISTOGRAM



BOX PLOT

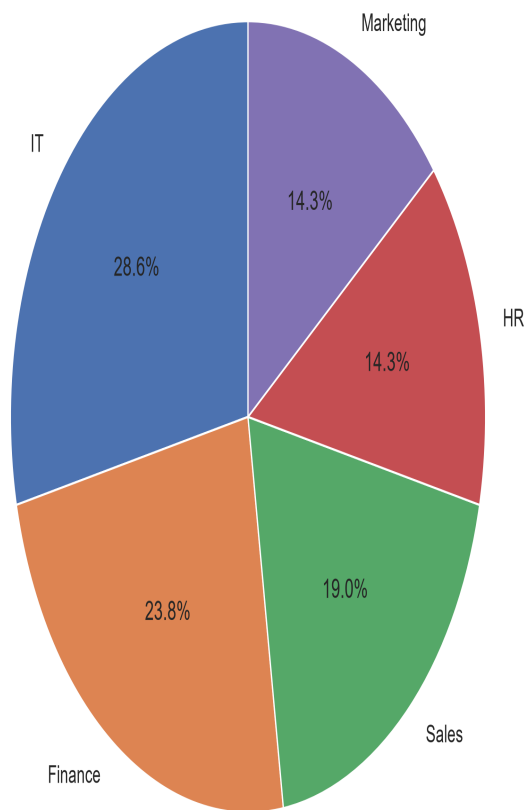


CORRELATION HEATMAP

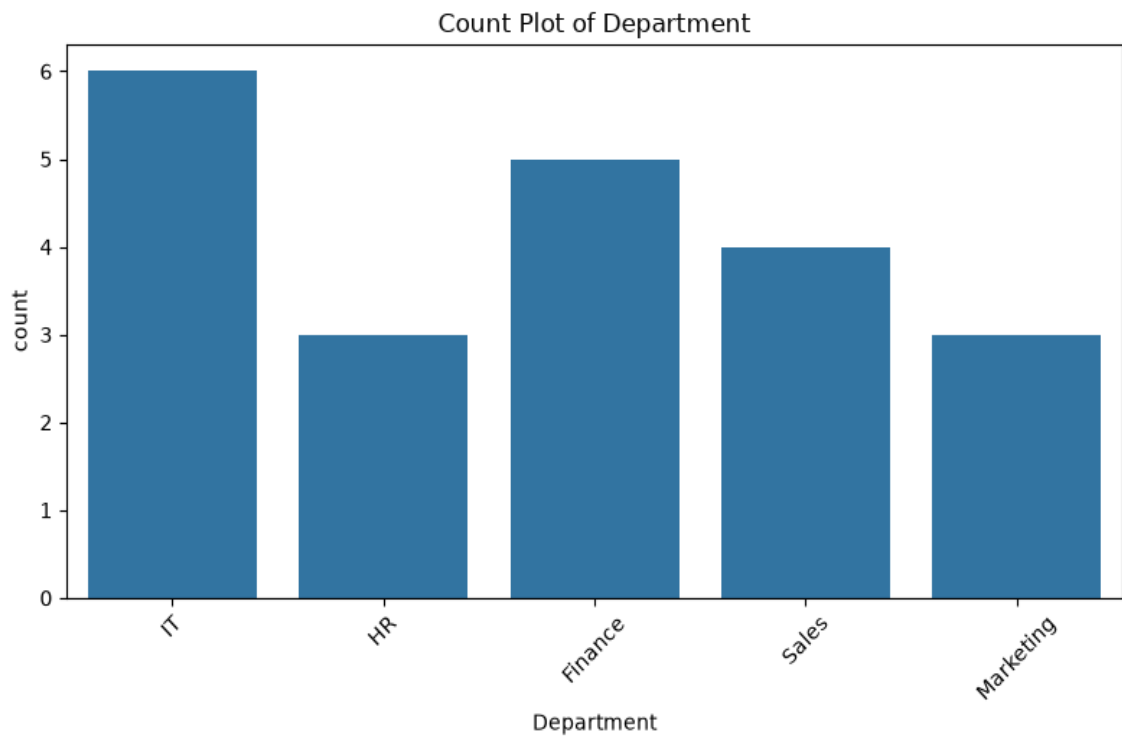


PIE CHART

Department Distribution



COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Experience	1
Salary	0
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

AI RECOMMENDATIONS

1. Execute Data Cleaning:

Remove the 1 duplicate row to prevent analytical skewing.

Impute the 1 missing Salary value using a regression model based on the employee's `Age` and `Experience` (justified by the 0.99 correlation), or manually reconcile it with payroll records.

Inspect and validate the 1 Experience outlier to ensure it aligns with the employee's biological age.

2. Audit the Performance Appraisal Process:

Introduce objective, key performance indicators (KPIs) and OKRs (Objectives and Key Results) to decouple performance ratings from age and tenure. Ensure junior employees have the structural path to achieve top-tier performance ratings.

3. Introduce Merit-Based Compensation Bands:

Transition from a purely seniority-based compensation structure to a hybrid structure that incorporates performance bonuses and merit-based salary bumps, allowing high-performing junior staff to be compensated competitively.

Rule-Based Recommendations

- ✓ ■■ Dataset contains 1 missing values. Consider filling or removing them.
- ✓ ■■ Dataset contains 1 duplicate rows. Consider removing them.
- ✓ ■■ 1 potential outliers detected.
- ✓ ■ Strong correlations detected. A Correlation Heatmap is recommended.
- ✓ ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ✓ ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ✓ ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.